

Kritarth Dandapat

Incoming PhD Student, Michigan State University | Undergraduate Researcher, University at Buffalo

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Professional Summary

Incoming PhD student in Computer Science at Michigan State University (Fall 2026), joining the Data Mining Laboratory under Dr. Pang-Ning Tan to work on spatiotemporal machine learning, deep learning-based weather forecasting, and AI adversarial robustness. Currently completing an accelerated three-year BS in Computer Science (Minor: Statistics) at the University at Buffalo, 3.8+ GPA.

Combines research and engineering: co-authored work in digital health and computational materials science, and has independently built and deployed production ML systems spanning mobile health, orthodontic monitoring, and telerehabilitation. Recognized with the PEARL undergraduate research award (\$2,500) and second place at the UB Health Futures Challenge (\$1,000).

Education

PhD in Computer Science

Starting Fall 2026

Michigan State University

East Lansing, MI

- Department of Computer Science and Engineering; Data Mining Laboratory (Advisor: [Dr. Pang-Ning Tan](#))
- Planned research focus: spatiotemporal machine learning, deep learning-based weather forecasting (DLWF), and AI adversarial robustness

Bachelor of Science in Computer Science, Minor in Statistics

August 2023 – June 2026

University at Buffalo, SUNY

Buffalo, NY

- Specialization in artificial intelligence; coursework spanning reinforcement learning, computer vision, machine learning, quantum computing, and distributed systems
- GPA 3.8+; Dean's List all semesters; accelerated three-year completion (21–22 credit semesters)
- Presidential Scholarship: \$15,000 per year; PEARL Award: \$2,500 competitive undergraduate research grant (Experiential Learning Network)

Research Experience

Research Assistant

June 2024 – Present

Embedded Sensing and Computing (ESC) Group, University at Buffalo

Buffalo, NY

Mobile sensing for healthcare and rehabilitation with [Prof. Wenyao Xu](#).

OralScan: AI-Assisted Geriatric Oral Screening

June 2024 – February 2025

- Architected a full-stack mobile health platform (React Native, React web) integrating end-to-end **YOLOv8** vision pipelines for guided intraoral image acquisition, multi-class dental disease screening, and automated tooth numbering across temporal scan sessions.
- Formulated a spatio-temporal scan-guidance system that aggregates sequential intraoral frames to produce consistent disease-detection signals robust to user-induced motion variability.
- Co-authored formative usability and acceptability study submitted to *Smart Health* (under review); delivered oral presentations to clinical stakeholders at CTSI and SURC 2024.
- Awarded second place (\$1,000), University at Buffalo Health Futures Challenge (Spring 2026).

OrthoScan: Orthodontic Remote Monitoring

February 2025 – October 2025

- Engineered mobile and backend systems for at-home orthodontic progress tracking, deploying **YOLO**-based hardware detection paired with depth-assisted spatial measurement pipelines to quantify structural change across longitudinal patient visits.
- Reconstructed and analyzed **3D dental structures** from intraoral scans using pre-trained **GNNs** for 3D

mesh processing; computed quantitative tooth position changes in both 2D image space and 3D world coordinates to track orthodontic progress with geometric precision.

- Translated sequential intraoral imagery into calibrated geometric progress signals, enabling quantitative remote monitoring between in-office appointments and reducing reliance on subjective clinician assessment.

mRehab: Telerehabilitation Platform (mrehab.agency)

November 2025 – Present

- Provisioned and configured **AWS** backend infrastructure to host testing and staging environments, enabling AI agent capabilities to operate within the live application stack.
- Deployed a real-time live sensor logging site for continuous ingest and monitoring of wearable sensor streams during active therapy sessions.
- Implemented **Kalman filters** for sensor fusion and activity detection, improving signal smoothness and robustness of exercise recognition under noisy real-world IMU conditions.
- Contributed to session logging, multi-modal sensor ingest pipelines, and exercise scoring algorithms deployed in live outpatient therapy workflows.

Undergraduate Researcher

June 2025 – Present

Peng Research Lab, University at Buffalo

Buffalo, NY

Computational materials science and clean-energy applications with [Prof. Jiayu Peng](#).

Perovskite Ordering and Symmetry-Aware Graph Neural Networks

June 2025 – November 2025

- Benchmarked symmetry-aware GNN architectures (including **ALIGNN**) on ordering-dependent energetics in crystalline perovskites; contributed benchmarking results and revised analysis to arXiv:2409.13851 and the open-source `PerovskiteOrderingGCNNs` repository.
- Migrated experiment tracking infrastructure from SigOpt to **Weights & Biases**; automated training loops, hyperparameter sweeps, and multi-architecture evaluation pipelines, reducing manual overhead and improving reproducibility.
- Co-authored commentary on agentic AI for multimetallic catalyst discovery, published on *ChemRxiv* (2025).

Equivariant MLIPs and Alchemical Extensions

December 2025 – Present

- Extended a dual-topology alchemical graph construction – originally built for **MACE** – to two additional equivariant MLIP architectures (fairchem **eSEN**, Meta **UMA**), enabling differentiable composition interpolation over disordered structures without retraining.
- Designed a λ -weighted message-passing and expert-routing scheme for UMA’s **Mixture-of-Linear-Experts** backbone, correcting double-counting in composition-dependent expert gating and adding an endpoint-collapse mechanism for exact pure-composition limits.
- Discovered and fixed a reference-energy bug causing non-physical ~ 15 eV energy discontinuities near pure compositions – invisible to force-based relaxation but critical for free-energy calculations – and added regression tests to prevent recurrence.
- Built a 12+ test automated verification suite (energy/force/stress/gradient agreement to $\sim 10^{-7}$ eV) plus benchmark figures and formula-level documentation cross-validating **MACE**, **eSEN**, and **UMA**.

Publications and Preprints

- **Dandapat, K.**, Emily [Surname], Master, T. A., Das, A., Bo, W., Cavuoto, L., and Xu, W. (2026). From Community Feedback to Measurement Redesign: Iterating mRehab for Accessible Home-Based Stroke Rehabilitation. *HumanSys 2026*. Preprint.
- **Dandapat, K.**, Master, T. A., Das, A., Bo, W., Lei, M., Cavuoto, L., Subryan, H., and Xu, W. (2026). Separating Safety from Preference: A Role-Gated Evaluation Platform for Clinical Rehabilitation LLMs. *HumanSys 2026*. Preprint.
- Lei, M., Das, A., Master, T. A., **Dandapat, K.**, Reddipogu, P., Xian, J., Rowe, V., Craft, L., Tabb, K., Cavuoto, L., Bhattacharjya, S., Jo, H. J., Subryan, H., Bo, W., and Xu, W. (2026). Towards AI Agents for Intelligent Voice-Driven Interaction in a Home-Based Stroke Rehabilitation System. University at Buffalo and Georgia State University. Preprint.

- **Dandapat, K.**, et al. (2026). Demo: mRehab – A Tangible, Offline-First Mobile Platform for Post-Stroke Upper-Limb Rehabilitation with an Edge-Cloud Voice [Assistant]. *MobiComm 2026* (Demo Track). Preprint.
- Soni, P., **Dandapat, K.**, Gherardi, A., Bo, W., Li, R., and Xu, W. (2025). OralScan, an AI-Powered Mobile Tool for Geriatric Oral Healthcare: A Formative Usability and Acceptability Study. Submitted to *Smart Health* (under review).
- Peng, J., Liu, C., Luo, Y., and **Dandapat, K.** (2025). Accelerating Multimetallic Catalyst Discovery with Robotics and Agentic AI. *ChemRxiv*, version 1. DOI: [10.26434/chemrxiv-2025-13n3f](https://doi.org/10.26434/chemrxiv-2025-13n3f).
- Peng, J., et al. (2024). Learning Ordering in Crystalline Materials with Symmetry-Aware Graph Neural Networks. *arXiv* preprint [arXiv:2409.13851](https://arxiv.org/abs/2409.13851). (Contributed benchmarking and revisions.)

Research Presentations

- Oral presentation and live demo – CTSI Research Group of Doctors, Nurses, and Medical Students (April 2024)
- Oral presentation – SUNY Undergraduate Research Conference (SURC) (April 2024)

Selected Projects

PaddockPsychRL [\[GitHub\]](#)

Multi-Agent RL

- Formulated a multi-agent RL framework integrating Formula 1 temporal performance signals into agent psychology models within **PettingZoo** cooperative environments, mapping real-world behavioral dynamics to learned policy structure.
- Constructed driver-specific agent profiles from **FastF1** 2025 lap-time consistency metrics, encoding temporal performance variability as per-agent action-noise scales to reflect authentic decision-making under pressure.
- Benchmarked tabular Q-learning and **MAPPO** baselines against psych-conditioned policies; psych-aware agents improved Strategy Resilience Score from 0.54 to 0.63 (+16.7%) under noisy multi-agent coordination.

Stack: Python, FastF1, PettingZoo, Ray **RLlib**, MAPPO, Q-learning

Other Projects

Computer Vision

- **Marine Guardian** – Geometric classifier + MobileNetV2 pipeline for ship detection in satellite imagery; 98.72% classification accuracy, benchmarked against EfficientNet and ResNet-50 baselines.
- **Density-Based Crowd Counting** – CSRNet trained in PyTorch for per-pixel density estimation in highly occluded surveillance scenes, avoiding failure modes of bounding-box detectors.
- **Facial Expression Benchmark** – Controlled comparison of CNN, ResNet-34, and ViT backbones for emotion recognition; 87.5% top-1 accuracy on the best configuration.

Professional Experience

Teaching Assistant, CSE 341: Computer Architecture

January 2026 – May 2026

Department of Computer Science and Engineering, University at Buffalo

Buffalo, NY

- Conducted weekly tutorial sessions on **Assembly** and **VHDL** for 30+ students, live-coding solutions to architecture problems including cache design, memory hierarchies, and instruction pipelining.
- Led recitations walking through processor pipeline stages, hazard detection, and cache replacement policies; developed worked examples and problem sets to reinforce lecture material.
- Held weekly office hours; graded exams and assignments; supported 100+ students asynchronously on Piazza throughout the semester.

Tutor and Peer-Assisted Leader (PAL)

August 2024 – December 2025

Tutoring and Academic Support Services (TASS), University at Buffalo

Buffalo, NY

- Served as **PAL for STA 119** (Introduction to Statistics): designed weekly worksheets aligned to recent lectures, then led two 50-minute collaborative sessions per week solving and explaining problems with students; course feedback indicated approximately 30% improvement in quiz performance.

- Provided one-to-one 1-hour tutoring appointments across seven courses: **STA 119, CSE 115, CSE 116, CSE 220, MTH 141, MTH 142**, covering introductory programming, data structures, and calculus.

Founder, Inference Foundry

Open Research Software Initiative

May 2026 – Present

Remote

- Launched an open-source initiative building reproducible ML inference tooling and shared contributor infrastructure for the scientific ML community (inference-foundry.rweb.site).

Founder and Vice President

UB National Student Data Corps (NSDC)

October 2023 – May 2024

Buffalo, NY

- Co-founded the UB chapter; launched the chapter website, organized data science workshops, and built networking programs connecting undergraduates to research and industry opportunities.

Honors and Awards

- **PEARL Award** – \$2,500 competitive undergraduate research grant, University at Buffalo Experiential Learning Network (November 2025)
- **Health Futures Challenge** – Second place (\$1,000) for OralScan, University at Buffalo (Spring 2026)
- **Cybersecurity Excellence** – Top 100 globally, Northeastern Cybersecurity C2C Finals; 10th in world finals (2025)
- **Collegiate Lockdown** – Top two teams representing UB; 4th in finals (2025)
- **Dean's List** – University at Buffalo (Fall 2023, Spring 2024, Fall 2024)
- **Presidential Scholarship** – \$15,000 per year

Competitions and Challenges

- IAA AppXcelerate Application, University at Buffalo (January–March 2025) – OralScan
- Russell L. Agrusa CSE Student Innovation Competition (November 2024) – team member, OralScan
- Aging Innovations Challenge (November 2024) – team member, OralScan
- Community Champions for Disability Health Challenge (October 2024) – OralScan

Technical Skills

- **Languages:** Python, JavaScript, Java, C++, Rust
- **Machine learning and AI:** PyTorch, TensorFlow, computer vision, deep learning, graph neural networks (GNNs), YOLOv8, Vision Transformer (ViT), reinforcement learning (MAPPO, Q-learning), Ray RLlib
- **Scientific ML and materials:** ALIGNN, machine learning interatomic potentials (MLIPs), MACE, E(3)-equivariant MPNNs, Weights & Biases, PyTorch Geometric
- **Software engineering:** React, React Native, Node.js, Django, Git, CUDA, OpenCV, NumPy, Matplotlib
- **Databases and cloud:** MongoDB, SQL, Firebase, SQLite, AWS, Google Colab, Firebase Hosting

Research Interests

- Spatiotemporal machine learning and deep learning-based weather forecasting (DLWF)
- AI adversarial robustness and out-of-distribution generalization
- Geometric and equivariant deep learning; graph neural networks for structured data
- Computer vision and sequence modeling for healthcare and scientific applications
- Machine learning for materials science and scientific discovery